

TREE-01

Q8)

Question:

DS	Session	Tree 1	Question Information	Level 1	Challenge 68
Question description					
A company has n employees with certain salaries. Your task is to keep track of the salaries and process queries.					
Constraints					
• $1 \leq n, q \leq 2 \cdot 10^5$ • $1 \leq p_i \leq 10^9$ • $1 \leq k \leq n$ • $1 \leq x \leq 10^9$ • $1 \leq a \leq b \leq 10^9$					
Input					
The first input line contains two integers n and q : the number of employees and queries. The employees are numbered $1, 2, \dots, n$.					
The next line has n integers $p_1, p_2, \dots, p_n$: each employee's salary.					
After this, there are q lines describing the queries. Each line has one of the following forms:					
• ! k x: change the salary of employee k to x • ? a b: count the number of employees whose salary is between $a \dots b$					

PROGRAM:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  typedef struct {
5      char t;
6      int a,b;
7  } Q;
8
9  int cmp(const void *a,const void *b){
10     return *(int*)a-*(int*)b;
11 }
12
13 int lower(int *a,int n,int x){
14     int l=0,r=n;
15     while(l<r){
16         int m=(l+r)/2;
17         if(a[m]<x) l=m+1;
18         else r=m;
19     }
20     return l;
21 }
22
23 void add(int *bit,int n,int i,int v){
24     for(;i<n;i+=i&-i) bit[i]+=v;
25 }
26
27 int sum(int *bit,int i){
28     int s=0;
29     for(;i;i&=i-1) s+=bit[i];
30     return s;
31 }
```

```
32
33 int main(){
34     int n,q;
35     scanf("%d%d",&n,&q);
36
37     int *p=malloc(n*sizeof(int));
38     int *all=malloc((n+q)*sizeof(int));
39     Q *qu=malloc(q*sizeof(Q));
40
41     for(int i=0;i<n;i++){
42         scanf("%d",&p[i]);
43         all[i]=p[i];
44     }
45
46     int m=n;
47
48     for(int i=0;i<q;i++){
49         scanf(" %c%d%d",&qu[i].t,&qu[i].a,&qu[i].b);
50         if(qu[i].t=='!')
51             all[m++]=qu[i].b;
52     }
53
54     qsort(all,m,sizeof(int),cmp);
55
56     int k=0;
57     for(int i=0;i<m;i++)
58         if(i==0||all[i]!=all[i-1])
59             all[k++]=all[i];
60
61     int *bit=calloc(k+1,sizeof(int));
62
63     for(int i=0;i<n;i++)
64         add(bit,k,lower(all,k,p[i])+1,1);
65
66     for(int i=0;i<q;i++){
67         if(qu[i].t=='!'){
68             int id=qu[i].a-1;
69             add(bit,k,lower(all,k,p[id])+1,-1);
70             p[id]=qu[i].b;
71             add(bit,k,lower(all,k,p[id])+1,1);
72         }else{
73             int l=lower(all,k,qu[i].a);
74             int r=lower(all,k,qu[i].b+1);
75             printf("%d\n",sum(bit,r)-sum(bit,l));
76         }
77     }
78
79     return 0;
80 }
```

OUTPUT:

TEST CASE 1:

```
5 3
3 7 2 2 5
? 2 3
! 3 6
? 2 3
3
2
```

TEST CASE 2:

```
5 3
31 17 12 12 15
? 12 31
! 13 6
? 2 13
5
3
```